=== UNIT 2: THEORY AND ENGINEERING APPLICATIONS OF FRICTION ===

>> 2.1 Coulomb’s Laws of Dry Friction

Friction is the tangential retarding force developed at the interface of two contacting bodies that resists impending or actual relative sliding motion. In civil engineering, friction is indispensable in foundation stability, retaining wall design, slope mechanics, and structural bolting.
* Static vs Kinetic Friction: (1) Static Friction (F_s): Tangential force resisting relative motion when bodies are at rest; it is self-adjusting from zero up to a maximum threshold known as Limiting Friction (F_lim). (2) Kinetic Friction (F_k): Resisting force during actual relative sliding, which is strictly less than limiting friction and independent of sliding velocity over moderate ranges.
* Coulomb’s Laws of Friction: (1) Limiting frictional force F_lim is directly proportional to the normal reaction N: F_lim = mu_s * N. (2) Kinetic friction F_k is directly proportional to normal reaction: F_k = mu_k * N, with mu_k < mu_s. (3) Frictional force is independent of the nominal apparent macroscopic contact area. (4) Frictional force depends fundamentally on the nature and roughness of contacting surface materials.

>> 2.2 Angle of Friction, Angle of Repose, and Cone of Friction

* Angle of Friction (phi): The angle between the resultant contact reaction R (vector sum of normal reaction N and limiting friction F_lim) and the normal to the contact plane: tan(phi) = F_lim / N = (mu_s * N) / N = mu_s. Thus: phi = atan(mu_s).
* Angle of Repose (alpha): The maximum inclination angle of a rough inclined plane with the horizontal at which a loose body placed on it will remain in static equilibrium without sliding down under its self-weight alone.
Proof that Angle of Repose equals Angle of Friction: For impending sliding down an incline of angle alpha: Resolving along incline: W * sin(alpha) = F_lim. Resolving perpendicular to incline: W * cos(alpha) = N. Dividing: tan(alpha) = F_lim / N = mu_s = tan(phi) => alpha = phi.
* Cone of Friction: If the resultant contact force R is revolved 360 degrees about the normal axis at the contact point, it sweeps out a right circular cone of apex angle 2*phi known as the Cone of Friction. If an applied force resultant falls within this cone, motion cannot occur regardless of force magnitude.

>> 2.3 Motion of Bodies on Rough Inclined Planes

Let a body of weight W rest on a plane inclined at angle theta to the horizontal, subjected to an external force P acting at angle beta to the incline:
* Case 1: Impending motion UP the plane: Friction force F acts DOWN the plane.

Sigma F_parallel = 0: P*cos(beta) - W*sin(theta) - mu*N = 0.
Sigma F_perp = 0: N + P*sin(beta) - W*cos(theta) = 0.
* Case 2: Impending motion DOWN the plane: Friction force F acts UP the plane.
Sigma F_parallel = 0: P*cos(beta) - W*sin(theta) + mu*N = 0.

>> 2.4 Ladder Friction and Tip-Slip Mechanics

A ladder leaning against a wall represents a classic structural equilibrium system under distributed and point live loads:
* Contact points: Floor A (reaction N_A upward, friction F_A inward toward wall) and Wall B (reaction N_B horizontal away from wall, friction F_B upward if wall is rough).
* Governing Equations: Sigma Fx = 0 => F_A = N_B; Sigma Fy = 0 => N_A + F_B = W_ladder + W_climber; Sigma M_A = 0 determines the limiting position of climber before slipping.

>> 2.5 Wedge Friction and Power Screws

* Wedge Mechanics: Wedges are inclined planes used to lift heavy structural equipment or split timber by converting horizontal thrust into large vertical uplifting forces.
FBD of each block and wedge must incorporate normal reactions and limiting friction forces opposed to impending relative movement.
* Screw Friction: Square-threaded and V-threaded power screws convert rotational torque into linear thrust.
Torque to raise load: T = W * r_mean * tan(alpha + phi), where alpha is screw helix angle: tan(alpha) = Lead / (pi * d_mean).
Torque to lower load: T = W * r_mean * tan(alpha - phi).
Self-Locking Condition: A screw will not back-drive under load if friction angle exceeds helix angle: phi >= alpha. Mechanical efficiency of self-locking screws is strictly below 50%.

>> 2.6 Belt and Rope Friction

* Flat Belt Transmission: Relates tight side tension T1 to slack side tension T2 over a curved pulley or capstan drum with wrap angle theta (in radians):
Formula: T1 / T2 = e^(mu * theta), where e = 2.71828...
* V-Belt Transmission: Account for groove wedge angle 2*beta: T1 / T2 = e^(mu * theta / sin(beta)).
* Power Transmitted by Belt: P = (T1 - T2) * v, where v = pi * D * N / 60 m/s.

[EXAMPLE 1: Block on Incline with Horizontal Force]

Problem Context: A concrete cube of weight W = 500 N rests on a rough ramp inclined at theta = 30 deg to the horizontal. Coefficient of static friction is mu_s = 0.35.
Determine the range of horizontal force P applied to the block such that it remains in static equilibrium (neither slides down nor slips up).
Solution Step 1: Case A - Impending motion DOWN the incline (minimum force P_min):
Friction acts UP the incline: F = mu_s * N.
Normal Reaction: N = W*cos(30) + P*sin(30) = 500*0.8660 + P*0.5 = 433.01 + 0.5*P.
Equilibrium along incline: P*cos(30) + mu_s * N - W*sin(30) = 0.
P*0.8660 + 0.35 * (433.01 + 0.5*P) - 500*0.50 = 0.
0.8660*P + 151.55 + 0.1750*P - 250 = 0 => 1.0410*P = 98.45 => P_min = 94.57 N.
Solution Step 2: Case B - Impending motion UP the incline (maximum force P_max):
Friction acts DOWN the incline: F = mu_s * N.
Normal Reaction: N = 433.01 + 0.5*P.
Equilibrium along incline: P*cos(30) - mu_s * N - W*sin(30) = 0.
P*0.8660 - 0.35 * (433.01 + 0.5*P) - 250 = 0.
0.8660*P - 151.55 - 0.1750*P - 250 = 0 => 0.6910*P = 401.55 => P_max = 581.11 N.
Conclusion: For static equilibrium, the horizontal force P must satisfy: 94.57 N <= P <= 581.11 N.

[EXAMPLE 2: Ladder Friction Analysis]

Problem Context: A uniform ladder of length L = 6.0 m and weight W_L = 300 N rests against a smooth vertical wall at an inclination angle of theta = 60 deg with the rough horizontal ground (mu_floor = 0.30). A structural inspector of weight W_p = 750 N climbs up the ladder. Calculate how far up along the ladder (distance x from the base) the inspector can safely climb before the ladder begins to slip.
Solution Step 1: Support reactions at base A and wall B:
At smooth wall B: Normal reaction N_B is perpendicular to wall (horizontal); friction F_B = 0.
At rough floor A: Vertical reaction N_A; horizontal friction F_A directed toward the wall.
Solution Step 2: Global Equilibrium equations:
Sigma Fy = 0 => N_A - W_L - W_p = 0 => N_A = 300 + 750 = 1050 N.
At impending slipping at base A: F_A = F_lim = mu_floor * N_A = 0.30 * 1050 = 315 N.
Sigma Fx = 0 => N_B - F_A = 0 => N_B = 315 N.
Solution Step 3: Moment Equilibrium about base point A (Sigma M_A = 0):
Clockwise moments from weights = Counterclockwise moment from wall reaction N_B:
[W_L * (L/2) * cos(60)] + [W_p * x * cos(60)] - [N_B * L * sin(60)] = 0.
[300 * 3.0 * 0.50] + [750 * x * 0.50] - [315 * 6.0 * 0.8660] = 0.
450 + 375 * x - 1636.74 = 0 => 375 * x = 1186.74 => x = 3.16 m.
Conclusion: The inspector can safely climb up to distance x = 3.16 m along the ladder before slip occurs.

[EXAMPLE 3: Wedge Friction to Elevate Heavy Foundation Pier]

Problem Context: A 15-degree wedge is driven horizontally under a machine foundation pier of weight W = 25 kN. The coefficient of friction between all contacting surfaces (wedge/ground, wedge/block, block/vertical guide) is mu = 0.25. Calculate the horizontal force P required to drive the wedge inward and raise the pier.

Solution Step 1: Determine friction angle phi:

phi = atan(mu) = atan(0.25) = 14.04 deg.

Wedge angle alpha = 15.0 deg.

Solution Step 2: Equilibrium of the vertical block (pier):

Forces on block: Weight W = 25 kN downward; vertical guide reaction R1 inclined at phi = 14.04 deg to normal; interface reaction R2 from wedge inclined at phi = 14.04 deg to normal of wedge face (total angle from vertical = alpha + phi = 15 + 14.04 = 29.04 deg).

Using Lami's or force polygon on block: R2 * cos(alpha + phi + phi) or vertical equilibrium:

Sigma Fx = 0 => R1*cos(phi) - R2*sin(alpha + phi) = 0 => R1 = R2 * sin(29.04) / cos(14.04) = 0.5003 * R2.

Sigma Fy = 0 => R2*cos(alpha + phi) - R1*sin(phi) - W = 0.

R2*cos(29.04) - (0.5003*R2)*sin(14.04) = 25 kN.

R2 * [0.8743 - 0.5003*0.2426] = 25 => R2 * [0.7529] = 25 => R2 = 33.20 kN.

Solution Step 3: Equilibrium of the horizontal wedge:

Forces on wedge: Horizontal force P; interface reaction R2 from block; floor reaction R3 inclined at phi = 14.04 deg to vertical normal.

Sigma Fx = 0 => P - R3*sin(phi) - R2*sin(alpha + phi) = 0 => P = R3*sin(14.04) + 33.20*sin(29.04).

Sigma Fy = 0 => R3*cos(phi) - R2*cos(alpha + phi) = 0 => R3 = 33.20 * cos(29.04) / cos(14.04) = 33.20 * (0.8743 / 0.9701) = 29.92 kN.

Now compute P: P = 29.92 * sin(14.04) + 33.20 * sin(29.04) = 29.92 * 0.2426 + 33.20 * 0.4854 = 7.26 + 16.11 = 23.37 kN.

Conclusion: A horizontal force P = 23.37 kN is required to drive the wedge and lift the foundation pier.

[EXAMPLE 4: Flat Belt Transmission and Power Delivery]

Problem Context: A flat leather belt connects an electric motor pulley (diameter D = 300 mm, rotating at N = 1200 RPM) to an exhaust fan in a civil ventilation system. The contact angle is theta = 165 deg (2.8798 radians) and coefficient of friction is mu = 0.30. If maximum allowable belt tension is T1 = 1500 N, calculate: (1) Slack side tension T2, (2) Linear belt speed v, (3) Maximum power in kW transmitted without slip.

Solution Step 1: Belt Tension Ratio using Capstan Equation:

T1 / T2 = e^(mu * theta) = e^(0.30 * 2.8798) = e^(0.8639) = 2.3725.

T2 = T1 / 2.3725 = 1500 / 2.3725 = 632.24 N.

Solution Step 2: Linear Belt Velocity v:
v = (pi * D * N) / 60 = (pi * 0.30 m * 1200) / 60 = 18.85 m/s.
Solution Step 3: Maximum Power Transmitted:
Effective driving tension = T1 - T2 = 1500 - 632.24 = 867.76 N.
Power P = (T1 - T2) * v = 867.76 N * 18.85 m/s = 16357.28 W = 16.36 kW.
Conclusion: Slack side tension T2 = 632.24 N; belt speed v = 18.85 m/s; maximum power transmitted is 16.36 kW.

>> 2.7 University Review & Examination Questions

* Q1: Deduce the relationship between coefficient of friction, angle of friction, and angle of repose.
Answer: Consider a body resting on an inclined plane whose angle of inclination alpha is gradually increased until the body just begins to slide under self-weight alone.
Resolving forces parallel to the plane: W*sin(alpha) = F_lim. Resolving perpendicular to plane: W*cos(alpha) = N. Dividing gives tan(alpha) = F_lim / N. By Coulomb's law, F_lim / N = mu_s. By definition of angle of friction phi, tan(phi) = F_lim / N = mu_s.
Therefore, tan(alpha) = tan(phi) = mu_s, proving that Angle of Repose equals Angle of Friction (alpha = phi).
* Q2: Explain the self-locking condition of a power screw and derive the expression for mechanical efficiency.
Answer: A power screw is self-locking if it maintains its load without unwinding when the turning torque is removed. The torque to lower a load is T = W * r_mean * tan(alpha - phi). If helix angle alpha <= friction angle phi, the required lowering torque is zero or positive, meaning the screw cannot unwind spontaneously. Mechanical efficiency is defined as work output over work input: eta = tan(alpha) / tan(alpha + phi). For self-locking (alpha <= phi), maximum efficiency occurs as alpha approaches phi, where eta < tan(phi) / tan(2*phi) = (1 - tan^2(phi)) / 2 < 50%. Hence, self-locking power screws always operate below 50% efficiency.

>> 2.8 Key Takeaways and Revision Points

* Friction is self-adjusting in the static regime up to limiting friction F_lim = mu_s * N.
* Angle of repose equals angle of friction (alpha = phi = atan(mu)).
* Ladder slip is governed by moment equilibrium about the contact base.
* Belt drive tension ratio follows the exponential law: T1/T2 = e^(mu*theta).